

Nagorik Sheba

AI-Powered Civic Complaint Platform

Week 5 Progress Report

Web Programming and Mobile App Development — Course Project
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1. Executive Summary

Week 5 focused on two major deliverables for the Nagorik Sheba platform: a full dark/light appearance system across the web application, and the design and implementation of the layered staff workflow requested by the project supervisor. Both features were carried from requirements analysis through implementation, database changes, API endpoints, UI, and end-to-end verification. Alongside feature work, repository activity was tracked across six branches, with automation added to visualize development velocity. Of the 45 tracked tasks on the project board, 30 are complete (67%), 6 are in progress (13%), and 9 remain to do (20%), reflecting steady, on-track progress toward the Week 6 milestones.

2. Task Board Snapshot

The project board was reviewed at the close of Week 5. The distribution of the 45 tracked tasks is summarized below.

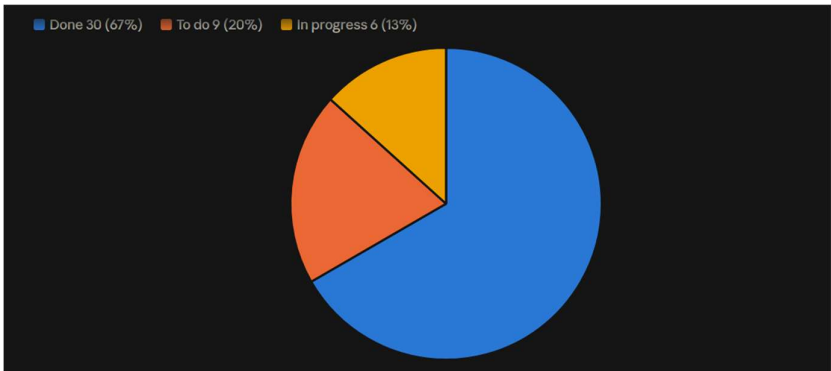


Figure 1: Task completion status (Week 5 snapshot)

Status	Task Count	Share
Done	30	67%
In Progress	6	13%
To Do	9	20%
Total	45	100%

Two-thirds of all planned tasks are now complete, and the in-progress items are concentrated in the SLA/auto-escalation and public analytics features scheduled for Week 6. No task is currently blocked.

3. Feature: Dark / Light Mode Theme System

Objective: Add a dark and light appearance option on the Landing page that instantly affects every other page, persists across sessions, and keeps all text readable in both modes.

3.1 Work Implemented

- **Theme infrastructure** — created `web/src/lib/theme.jsx` with a React `ThemeProvider` and `useTheme()` hook; the selected mode is persisted to `localStorage` (`ns_theme`) and applied as a light class on `<html>`, so switching modes instantly re-themes every page.
- **Token-based styling** — reworked `tailwind.config.js` so every color (backgrounds, panels, text, slate scale, accent hues, status colors) resolves through CSS variables with alpha support; `index.css` defines the dark palette as the `:root` default and a complete light palette under `html.light`, covering glass panels, buttons, inputs, chips, scrollbars, Leaflet map controls, and native `<select>` popups — re-theming the whole app without touching 100+ existing class names.
- **No-flash loading** — a pre-mount inline script in `web/index.html` applies the saved theme before React mounts, eliminating the flash of the wrong theme on load.
- **UI controls** — a prominent Appearance switcher on the Landing hero and a compact toggle in the Navbar on every page (desktop and mobile), fully bilingual (English/বাংলা).
- **Readability fixes** — dedicated on-accent text color for solid accent buttons (map layer switcher, Emergency directions), fixed the "Voted" button contrast, and darkened status hues (amber-700, rose-700, emerald-700) in light mode for WCAG-friendly contrast.

3.2 Verification

The production build passes cleanly, both dark and light token sets were confirmed present in the compiled CSS, and live testing confirmed the selected theme propagates instantly to every page and persists correctly across navigation and page reloads.

4. Feature: Layered Staff Workflow — Requirements Analysis & Design

Per the supervisor's instruction, verified complaints are routed through a hierarchical human review chain: Junior Staff → Senior/Mid Staff → Mayor/Chairman → Work Inspection Team (WIT) → Field Staff. The junior officer screens for fake or irrelevant complaints as a safety net above the AI's first-pass check; the senior officer assesses feasibility (manpower, budget, solvability); the mayor/chairman grants final permission; and the WIT assigns the field staff who execute the work. The system must expose the full per-complaint lifecycle, show which problem is under which staff member, provide department-wise views for staff and the Mayor/Chairman, and send email updates to both staff and citizens.

4.1 Approach Assessment

The proposed approach is realistic and mirrors how local government actually operates in Bangladesh (Union Parishad chairman / municipal mayor / city corporation mayor approval chains). It also fits the existing architecture well: the pre-existing `status_history`, `agent_logs`, and `notifications` structures already covered roughly 70% of the lifecycle foundation needed.

4.2 Key Design Decisions

- **Human layer over AI** — Agent-01 (Verifier) performs the first-pass spam/fake filter; human officers catch what the AI misses.
- **Department = complaint category** — road/electricity/water/gas/sanitation/other, already assigned by Agent-02 (Classifier), so department-wise views require no extra modeling.
- **Dedicated workflow_steps audit table** — (stage, action, actor, comment, timestamp) instead of overloading the single status field — this cleanly yields both the lifecycle chain and "which problem is under which staff."

- **Defined rejection paths** — at every layer with mandatory comments; the rejecting stage and reason are recorded and shown to the citizen.
- **Mayor permission flows back to senior staff** — for the WIT handoff, matching the supervisor's diagram (Mayor approves → mid staff → solving staff).
- **Voting lock timing** — voting stays open through the entire review chain and locks only once execution starts.
- **Email via Ethereum demo SMTP** — (nodemailer) requires no real credentials for demonstration; a production relay is a single environment variable away.
- **Mayor-bottleneck risk** — identified (every complaint waits on three human hops); flagged for mitigation via the SLA auto-escalation system planned for Week 6.

4.3 Designed Workflow

Five staff roles were defined — junior, senior, mayor, wit, field — governing the following status chain (with rejection possible at the junior, senior, and mayor stages):

Stage	Owner Role	Possible Outcome
Verified	System (AI Agent-01)	Forwarded to Junior Staff
Junior Review	Junior Staff	Approve → Senior, or Reject (with comment)
Senior Review	Senior/Mid Staff	Approve → Mayor, or Reject (with comment)
Mayor Review	Mayor / Chairman	Permission Granted → WIT, or Reject
WIT Review	Work Inspection Team	Assign Field Staff + ETA
In Process	Field Staff	Execute and upload resolution proof
Resolved	Citizen	Confirm or Dispute (auto-reopen after 3 days)

5. Feature: Layered Staff Workflow — Implementation & Verification

5.1 Backend (server/)

- **Workflow engine** — `utils/workflow.js` implements 10 guarded actions (`junior_approve/reject`, `senior_approve/reject`, `mayor_approve/reject`, `send_to_wit`, `wit_assign`, `field_start`, `field_resolve`), a central `setStatus()` transition chokepoint, and per-role action queues.
- **Database changes** — added a `staff_role` column (5 roles); a new `workflow_steps` audit table (`stage`, `action`, `actor name/role`, `comment`, `timestamp`); complaints gained `current_stage`, `stage_owner_role`, `executor_staff_id`, `executor_name`, `rejected_at_stage`, `resolution_image_url`, `resolved_confirmed_by_citizen`, and `confirmation_deadline`; the notifications table was rebuilt with `recipient_type` so staff also receive in-app notifications.
- **Endpoints** — a unified POST `/api/complaints/:id/workflow` (`approve/reject/assign/start/resolve`, multipart for the resolution photo), a role-filtered GET `/staff/queue`, a WIT-only GET `/staff/executors`, citizen POST `/:id/confirm` and POST `/:id/dispute`; the staff JWT now carries `staff_role`.
- **Email notifications** — `utils/emailer.js` (nodemailer) fires at every transition: new complaint → junior officer; each approval → next officer; permission granted → senior; assignment → field executor with ETA; every step → citizen; rejections → citizen and voters with the officer's comment.
- **Resolution proof & citizen confirmation** — field staff must attach a resolution photo when the complaint had supporting evidence (before/after rule, enforced server-side); the citizen then confirms or disputes — a dispute reopens the complaint into WIT re-inspection, and unconfirmed resolutions auto-reopen after a 3-day window via a 6-hourly system sweep (`autoReopenUnconfirmed`) that notifies WIT, senior, and mayor.

5.2 Web (web/src/)

- **LifecycleCard** — a public lifecycle timeline on every complaint: submission → AI verification (with department) → Junior Check → Senior Feasibility → Mayor/Chairman Permission → WIT → Assignment (executor + ETA) → Execution → Resolved; every step shows date, officer name, role badge, and comment; rejections render in red with stage and reason for full citizen transparency.
- **StaffDashboard** — role-based consoles for all 5 roles; senior and mayor get an oversight toggle ("My Queue" / "All Problems" full-lifecycle view); department-wise filtering plus a workflow-stage filter; stat counters; inline action buttons on each card.
- **WorkflowActions modal** — mandatory comment, WIT executor + ETA picker, resolution photo upload; ComplaintDetail gained the lifecycle card, resolution proof display, and citizen confirm/dispute banner.

5.3 Mobile (mobile/lib, Flutter)

A role-aware staff console with per-role status chips, a lifecycle timeline on the complaint detail screen, workflow actions, a WIT executor + ETA dialog, and full English/বাংলা localization.

5.4 Data & Verification

A 5-role staff hierarchy was seeded for every authority nationwide, following the pattern `{role}.{district}.{authorityId}@nagorik.bd` with password `staff123` (e.g. `mayor.dhaka.2@nagorik.bd`), along with demo complaints and backdated workflow chains. The full chain was tested end-to-end — junior → senior → mayor → WIT → field → resolve → citizen confirm — including rejection paths, voting-lock behavior, dispute-driven reopening, and email delivery verified through Ethereum preview links.

6. Repository & Branch Progress

Repository activity was reviewed across all branches to track development velocity going into Week 6.

6.1 Branch Overview

Branch	Commit	Status	Last Activity
0.1	ba60b638	Development	Version 1 foundation
0.2	dabb6fa8	Development	Week 2 additions
0.3	3cdae9ae	Development	Version 3 updates
0.4	4705e70e	Development	Version 4 updates
0.5	b216d4af	Development	Latest version release
automation/weekly-velocity-chart	d98f73d4	Active Feature	Current (in progress)

6.2 Recent Development Activity

- **Weekly velocity automation (Sept 11)** — a new automation/weekly-velocity-chart branch adds a Python script (using the GitHub REST API and matplotlib) that counts commits per version branch and renders them as a pie chart, giving an at-a-glance view of where development effort has concentrated across the 0.1–0.5 releases; the feature is under active review in PR #59.
- **Testing infrastructure (Aug 30)** — comprehensive unit tests added for authentication, the AI agent pipeline, and the database layer; package.json updated with test scripts and Vitest.
- **Documentation (Aug 23, Aug 30)** — README revised with project updates and features; implementation details and getting-started instructions documented.

- **Cleanup (Aug 20, Aug 30, Sept 11)** — removed a stale server directory, cleaned up unnecessary Python files, and reorganized the workflow directory.

6.3 Key Metrics

- Total tracked commits: 50+ across all branches
- Language composition: HTML 89.5% (frontend/UI focus), Python 10.5% (backend/automation tooling)
- Active development: ongoing, with the most recent commit within the last few hours of this report
- Open pull requests: 1 active (PR #59 — Weekly Velocity Chart automation)

Overall, the repository shows consistent iterative development: version-numbered branches (0.1 through 0.5) trace the platform's evolution, while the newer automation branch adds tooling to monitor that progress going forward.

7. Planned Work — Week 6

7.1 Resolution Proof & Citizen Confirmation (hardening)

- Field Staff uploads a before/after photo when marking a complaint resolved
- Citizen receives a "confirm resolved" prompt; no response or a dispute within N days triggers auto-reopen or escalation
- Add resolved_confirmed_by_citizen boolean to the COMPLAINT table and log disputes in COMPLAINT_TRAIL

7.2 SLA Timers and Auto-Escalation

- Maximum duration per workflow stage (e.g. 3 days at Junior, 5 days at Senior), scaled by priority score
- Scheduled job checks overdue complaints and auto-forwards them up the chain
- "Overdue" badge in staff console queues, sorted to the top

7.3 Public Accountability Analytics

- Per-authority dashboard: average resolution time, % resolved vs. rejected, volume by category
- Public leaderboard ranking local government institutions (LGIs) / authorities by responsiveness
- Heatmap overlay on the Leaflet map showing complaint density and resolution rate

8. Conclusion

Week 5 delivered two substantial, fully verified features — the dark/light theme system and the complete layered staff workflow — while also strengthening project visibility through repository tracking and the new velocity-monitoring automation. With 67% of planned tasks complete and no blockers, the project is on track to close out the remaining SLA-escalation and public-analytics work in Week 6, bringing the platform to a fully accountable, end-to-end civic complaint pipeline.